

# American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

## SE - L07 - DSDM Dynamic Systems Development Method

### Questions and Answers

Faculty: Mejbah Ahammad

Email: ahammadmejbah@aiub.edu

Semester: Summer 2025-2026

Date: July 10, 2026

#### 1. What is DSDM?

**Answer:**

Dynamic Systems Development Method (DSDM) is an Agile software development methodology based on Rapid Application Development (RAD). It focuses on rapid software delivery, continuous user involvement, iterative development, and flexibility in changing requirements.

#### 2. What is the main objective of DSDM?

**Answer:**

The main objective of DSDM is:

*“Deliver what the business needs when it needs it.”*

It aims to deliver high-value software quickly by focusing on important business requirements.

#### 3. Why is DSDM considered an Agile methodology?

**Answer:**

DSDM is considered Agile because it supports:

- Iterative and incremental development.
- Continuous customer collaboration.
- Flexible requirement changes.
- Frequent delivery of working software.

#### 4. What are the major phases of the DSDM process model?

**Answer:**

The major phases of DSDM are:

- (a) Feasibility Study
- (b) Business Study
- (c) Functional Model Iteration
- (d) Design and Build Iteration
- (e) Implementation

#### 5. What is the purpose of the Feasibility Study phase in DSDM?

**Answer:**

The Feasibility Study phase determines whether DSDM is suitable for a project by analyzing technical feasibility, organizational factors, risks, and project constraints.

Main outputs include:

- Feasibility Report
- Feasibility Prototype
- Development Plan
- Risk Log

**6. What is the Business Study phase in DSDM?**

**Answer:**

The Business Study phase analyzes business requirements and identifies important system characteristics.

It produces:

- Prioritized Requirements List
- Business Area Definition
- System Architecture Definition
- Prototype Plan

**7. What is Functional Model Iteration?**

**Answer:**

Functional Model Iteration is the DSDM phase where functional prototypes are developed, tested, and improved based on user feedback.

Activities include:

- Identify functional prototype.
- Create prototype.
- Review prototype.

**8. What is Design and Build Iteration?**

**Answer:**

Design and Build Iteration transforms functional prototypes into a tested software system.

It focuses on:

- Designing the system.
- Building prototypes.
- Testing functionality.
- Preparing user documentation.

**9. What happens during the Implementation phase of DSDM?**

**Answer:**

During implementation:

- Users approve the system.
- Users receive training.
- Final software is deployed.
- Business impact is reviewed.

**10. What is the 80/20 rule in DSDM?**

**Answer:**

The 80/20 rule states that approximately 80% of business value can be achieved by implementing the most important 20% of requirements.

It helps teams focus on essential functionality first.

**11. What is Timeboxing in DSDM?**

**Answer:**

Timeboxing is a technique where development work is completed within a fixed time period.

Characteristics:

- Fixed start and end dates.

- Usually 2–4 weeks duration.
- Focuses on essential requirements.

**12. Why is Timeboxing important?**

**Answer:**

Timeboxing is important because it:

- Controls project schedule.
- Prevents development delays.
- Improves resource management.
- Ensures regular delivery.

**13. What is MoSCoW prioritization?**

**Answer:**

MoSCoW is a requirement prioritization technique used in DSDM.

It categorizes requirements as:

- Must Have
- Should Have
- Could Have
- Won't Have this time

**14. Explain Must Have requirements in MoSCoW.**

**Answer:**

Must Have requirements are essential features that are necessary for project success.

Example:

Login system and payment processing in an online shopping application.

**15. Explain Should Have requirements in MoSCoW.**

**Answer:**

Should Have requirements are important features but the system can work temporarily without them.

Example:

Password recovery feature.

**16. Explain Could Have requirements in MoSCoW.**

**Answer:**

Could Have requirements are optional features that improve user experience but are not critical.

Example:

Dark mode or additional themes.

**17. What are Won't Have requirements?**

**Answer:**

Won't Have requirements are features that are not included in the current release but may be considered for future development.

**18. Why is prototyping used in DSDM?**

**Answer:**

Prototyping is used because:

- Users can validate requirements.
- Developers receive early feedback.

- Errors are detected early.
- Development confidence increases.

**19. Write two advantages of DSDM.**

**Answer:**

Two advantages of DSDM are:

- (a) Faster software delivery through iterations.
- (b) Better handling of changing requirements.

**20. Write two limitations of DSDM.**

**Answer:**

Two limitations of DSDM are:

- (a) Requires continuous user involvement.
- (b) Requires experienced Agile development teams.

**21. Why does DSDM keep time and resources fixed?**

**Answer:**

DSDM keeps time and resources fixed to ensure predictable project delivery. Instead of increasing cost or delaying schedules, lower priority requirements are adjusted or moved to future iterations.

**22. What is the key difference between Traditional Development and DSDM?**

**Answer:**

Traditional development fixes requirements while allowing time and resources to vary.

DSDM fixes time and resources while allowing functionality to vary according to priority.